

1.

Data size	Configuration	Training error	Validation error	Time of execution
1000	1 hidden layer 4 nodes	0.191	0.186674	144.39
10000	1 hidden layer 4 nodes	0.00360	0.003868	132.4244
100000	1 hidden layer 4 nodes	0.00158	0.001270	162.603471
1000	2 hidden layers of 4 nodes each	0.30600	0.294971	165.765
10000	2 hidden layers of 4 nodes each	0.00540	0.004873	121.507
100000	2 hidden layers of 4 nodes each	0.00173	0.002644	178.7459

2.

The model utilizing 100,000 data points and one hidden layer with four nodes demonstrates the best performance according to the provided results. The model configuration with 100,000 data points and 1 hidden layer with 4 nodes demonstrated the best generalization capability by reaching a validation error of 0.001270. The model using 100,000 data points and two network layers reached a similar training error rate (0.00173 vs 0.00158) but showed increased validation error of 0.002644. The single-layer model executes faster than its 2-layer counterpart because it requires less time (162.60s) compared to (178.75s). The selected model architecture strikes a perfect equilibrium between keeping data processing efficient while maintaining strong generalization ability and model simpleness.

3.

Method used	Dataset size	Testing-set predictive performance	Time taken for the model to be fit
XGBoost in Python via scikit-learn and 5-fold CV	1000	0.945	0.404
	10000	0.972	1.631
	100000	0.984	6.325

The results indicate that XGBoost outperforms deep learning techniques through all three data sets sizes. XGBoost delivered superior testing-set predictive results of 0.945, 0.972, and 0.984 for 1000, 10000, and 100000 observations respectively than the best deep learning model which demonstrated a validation error of 0.001270 equivalent to approximately 0.999 accuracy. XGBoost achieved remarkable computational efficiency during training by completing the process in 0.404 seconds, 1.631 seconds and 6.325 seconds while deep learning models needed more than 120 seconds in all configurations. XGBoost achieved superior performance on datasets containing 1000 observations by maintaining high predictive power (0.945) while deep learning models reached only 0.186674 as their best validation error. The ensemble model structure and generalization properties of XGBoost make it outperform neural networks on this problem space particularly in situations where available training data is restricted.